

Assignment = 1

I] Row-major and Column-major Address calculation

Array : `int A[4][5]`

Base Address = 1000

Size of int = 2 bytes

Row-major order: $A[i][j] = BA + ((i \times \text{No. of columns}) + j) \times \text{size}$

Column-major order: $A[i][j] = BA + ((j \times \text{No. of Rows}) + i) \times \text{size}$

i] `A[1][3]`

$$\text{Row-major} = 1000 + [(1 \times 5) + 3] \times 2 = 1016$$

$$\text{Column-major} = 1000 + [(3 \times 4) + 1] \times 2 = 1026$$

ii] `A[2][4]`

$$\text{Row-major} = 1000 + [(2 \times 5) + 4] \times 2 = 1028$$

$$\text{Column-major} = 1000 + [(4 \times 4) + 2] \times 2 = 1036$$

iii] `A[3][1]`

$$\text{Row-major} = 1000 + [(3 \times 5) + 1] \times 2 = 1032$$

$$\text{Column-major} = 1000 + [(1 \times 4) + 3] \times 2 = 1014$$

iv] `A[0][2]`

$$\text{Row-major} = 1000 + [(0 \times 5) + 2] \times 2 = 1004$$

$$\text{Column-major} = 1000 + [(2 \times 4) + 0] \times 2 = 1016$$

2) Sparse matrix Representation:

vij Total elements : $4 \times 5 = 20$

Total non-zero elements : 7

zero elements : 13

since, the no. of zero elements is greater than the size of non zero elements. it is a sparse matrix.

vij 3-Tuple Representation

Row	Column	Value
4	5	7
0	2	5
0	4	8
1	0	7
2	3	6
3	0	9
3	2	4
3	4	2

viii) memory Required

a-normal matrix Representation

$$20 \times 2 = 40 \text{ bytes}$$

b - 3-TUPLE Representation

Row 5 in tuple, $7+1 = 8$

Each row has 3 integers

Total, $8 \times 3 = 24$

memory : $24 \times 2 = 48$ bytes

ix] Assume the size of int as 2 bytes

Normal Matrix rep : $20 \times 2 = 40$ bytes

3-TUPLE rep : $8 \times 3 \times 2 = 48$ bytes.